

Syntax - Introduction

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Week 5: Lecture 1

What is Syntax?

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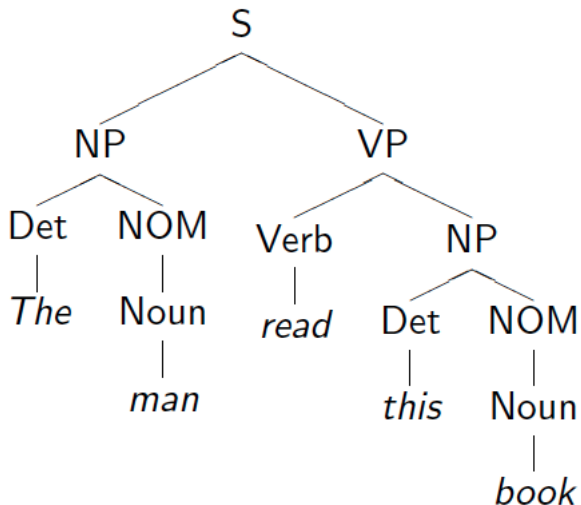
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- More complex notions: constituency, grammatical relations, subcategorization etc.

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Syntax Tree: Example



Defining the notions: Constituency

Constituent

A group of words acts as a single unit - phrases, clauses etc.

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Constituency: Noun Phrase

- *Kermit the frog*
- *they*
- *December twenty-sixth*
- *the reason he is running for president*

Constituent Phrases

Usually named based on the word that heads the constituent:

<i>the man from Amherst</i>	is a Noun Phrase (NP) because the head <i>man</i> is a noun
<i>extremely clever</i>	is an Adjective Phrase (AP) because the head <i>clever</i> is an adjective
<i>down the river</i>	is a Prepositional Phrase (PP) because the head <i>down</i> is a preposition
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Joe appears in a place that a larger noun phrase could have been.

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Kermit the frog comes on stage

They come to Massachusetts every summer

December twenty-sixth comes after Christmas

The reason he is running for president comes out only now.

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Constituent = Prepositional phrase: On December twenty-sixth

On December twenty-sixth I'd like to fly to Florida.

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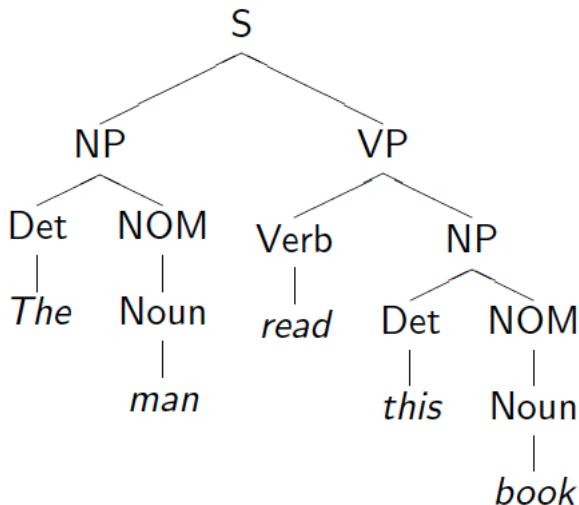
I'd like to fly to Florida on December twenty-sixth.

But not split apart

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Modeling Constituency: what tool do we need?



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The most common way of modeling constituency

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Noun phrase can be composed of either a ProperNoun or a determiner (Det) followed by a Nominal; a Nominal can be more than one nouns

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Noun phrase can be composed of either a ProperNoun or a determiner (Det) followed by a Nominal; a Nominal can be more than one nouns

$NP \rightarrow \text{Det Nominal}$

$NP \rightarrow \text{ProperNoun}$

$\text{Nominal} \rightarrow \text{Noun} \mid \text{Noun Nominal}$

CFG for Languages

CFG: $G = (T, N, S, R)$

- T : set of terminals
- N : set of non-terminals
 - ▶ For NLP, we distinguish out a set $P \subset N$ of pre-terminals, which always rewrite as terminals
- S : start symbol
- R : Rules/productions of the form $X \rightarrow \gamma$, $X \in N$ and $\gamma \in (T \cup N)^*$

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Terminals mainly correspond to words in the language while pre-terminals mainly correspond to POS categories

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Can you identify the terminal, non-terminals and preterminals?

CFG as a generator

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- Thus a CFG can be used to randomly generate a series of strings
- This sequence of rule expansions is called a derivation of the string of words, usually represented as a tree

A CFG defines a formal language = set of all sentences (string of words) that can be derived by the grammar

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- Sentences in this set are said to be **grammatical**
- Sentences outside this set are said to be **ungrammatical**

Recursive Definition

- $PP \rightarrow \text{Prep NP}$
- $NP \rightarrow \text{Noun PP}$

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Example Sentence

$[_S \text{The mailman ate his } [_{NP} \text{lunch } [_{PP} \text{with his friend } [_{PP} \text{from the cleaning staff } [_{PP} \text{of the building } [_{PP} \text{at the intersection } [_{PP} \text{on the north end } [_{PP} \text{of town}}]]]]]]]$.

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$A \rightarrow BC$

- I can rewrite A as B followed by C regardless of the context in which A is found
- Or when I see a B followed by a C , I can infer an A regardless of the surrounding context